

Why do marked words persist? Corpus and experimental evidence for competition between phonological and semantic well-formedness in word-formation and the lexicon

Introduction. From a probabilistic grammar perspective, grammatically marked forms can be underrepresented but nevertheless tolerated (Kempen & Harbusch, 2008; Martin, 2011; Breiss & Hayes, 2020, *i. a.*). This study examines why marked forms persist in the lexicon rather than disappearing over time (Martin, 2011). We focus on phonologically marked compounds in English, Finnish, and Hungarian and propose that *word form flexibility* (henceforth, flexibility), measured as the number of alternative forms compatible with the target semantics, competes with phonological well-formedness to determine what form enters the lexicon. We propose that marked forms survive because there are fewer viable semantic alternatives available for speakers to use. While intuitive, no study has systematically examined this claim. We present corpus studies and a nonce compound formation experiment. Both yield results in support of our hypothesis.

English, Finnish, and Hungarian corpus studies. We first examined whether attested phonologically marked compounds are more likely to have lower flexibility in English, Finnish, and Hungarian. The markedness constraints investigated were *GEMINATE in English (e.g., *bookkeeper*; Martin, 2011) and *vowel backness harmony* in Finnish (e.g., *kesälomaa* “summer vacation”; Duncan, 2015) and Hungarian (e.g., *kőpor* “rock flour”; Siptár & Törkenczy, 2000). We collected two-member compounds in these languages from Wiktionary and published datasets (Gagné *et al.*, 2019; Günther & Marelli, 2019). First, we found compounds containing marked structures (geminate, disharmonic members) were underrepresented by a chi-squared test, suggesting they are indeed lexically disfavored (Figure 1). Second, each compound’s flexibility was calculated as the product of the summed cosine similarities between each member and all the other words in a language-specific word2vec model. This metric served to indicate the semantic similarity of the compound across the lexicon in the language. In each language, a linear mixed effects model found that the probability of a compound being marked increased as its flexibility decreased (Figure 2), suggesting that flexibility likely modulates the extent of phonological optimization in the lexicon.

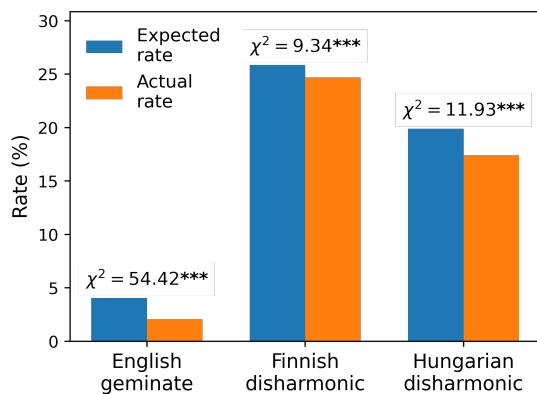


Figure 1. The underrepresentation of geminate and disharmonic compounds in the three languages.



Figure 2. Corpus study results: the model-derived interaction between a compound’s word form flexibility and the markedness rate.

English nonce compound formation experiment. To gather complementary evidence for the real-time competition of phonological markedness and semantic flexibility in word-formation, we conducted two experiments examining English nonce compound formation, where participants were instructed to form nonce compounds given combinations of two noun concepts (cf. Figure 3). In order to estimate the relationship between the flexibility of a noun concept, vocabulary size, and education level, we first conducted a norming study. 80 American

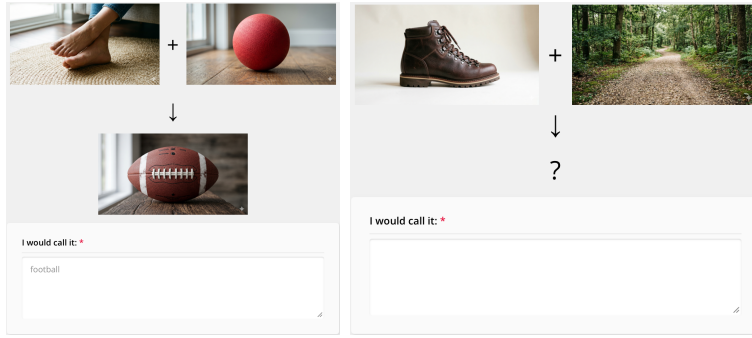


Figure 3. Example trials in the compound formation experiment. The trials prompt the participant to create a real compound (left) and a nonce compound (right).

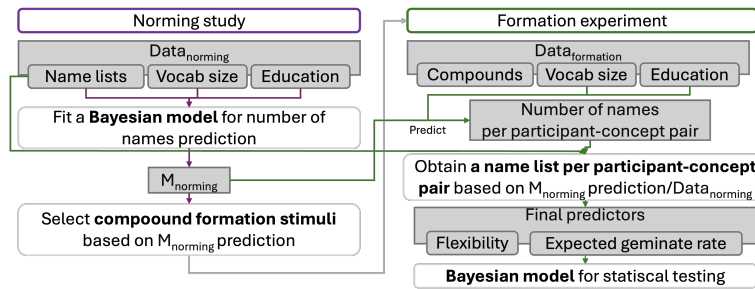


Figure 4. Experimental flow and data processing.

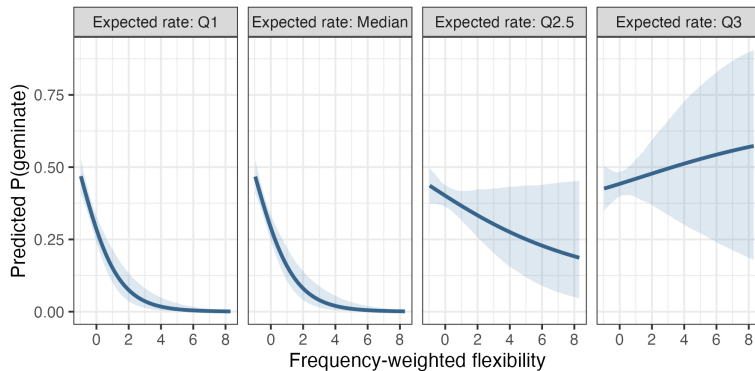


Figure 5. Compound formation experiment results: the model-derived interaction between a compound's flexibility and the markedness rate of the name offered by the participant.

distributions in the norming study. Flexibility was then calculated as the number of names, weighted by their frequencies in the norming data. To account for the baseline geminate rate, the expected geminate rate for each participant-concept pair was also estimated. The results of a Bayesian linear mixed effects model (plotted in Figure 5) showed a significant negative correlation between flexibility and geminate rate. This suggests that when a novel concept pair has a lower flexibility, the participant is more likely to violate the phonological constraint in the name they offer. A competing effect from expected geminate rate also surfaced, where the flexibility effect weakened at extremely high expected geminate rates, suggesting a “ceiling.”

Discussion Both the corpus and experimental results supported the hypothesis that flexibility competes with phonological well-formedness in marked forms. A compound is more likely to violate the markedness constraint when the language user has fewer alternatives to express the target meaning. This suggests that semantic flexibility is a moderating constraint on phonological avoidance of marked structures that shapes both static lexical distribution and online language use.

English speakers offered as many names as possible for pictures of 80 noun concepts selected to vary in their number of word forms based on WordNet (Miller, 1995). Participants also provided demographic information and completed a *MINT sprint* vocabulary size assessment (Garcia & Gollan, 2022). A Bayesian model was fit to predict the number of names from concept identity, education level, and vocabulary size. Concepts were then classified as high- or low-flexibility and paired to maximize expected geminate rates in novel compounds. We obtained 20 high- and 20 low-flexibility concept pairs. In the formation experiment, a separate group of 86 American English speakers named these concept pairs, after being familiarized with the process with real compounds, such as *football*. Demographic and vocabulary size were also collected. To best estimate the specific names that were available for each participant when seeing a concept, for each participant-concept pair, we predicted a list of names based on the norming model and the concepts' name